

Total number of pages: 1
Degree: B.Tech

Roll no.:
Semester: Summer

Summer Semester Examination, July 2026

Course Code: ITMTC04/ITMTC301

Course title: Optimization Principles and Techniques

Maximum Marks. 50

Time: 3 hours

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Note: Missing data/information (if any), May be used		Marks	CO																																																	
Q. No.	Questions																																																			
Q1	Attempt any 2 parts of the followings																																																			
1a	Solve the following LPP by the two-phase simplex method and verify your answer graphically. Max $Z = 4x_1 + 3x_2$ subject to $x_1 + x_2 \leq 8$ $2x_1 + x_2 \geq 10; x_1, x_2 \geq 0.$	5	CO1																																																	
1b	A firm manufactures two products A & B on which the profits earned per unit are Rs. 3 and Rs. 4 respectively. Each product is processed on two machines M_1 and M_2 . Product A requires one minute of processing time on M_1 and two minutes on M_2 , while B requires one minute on M_1 and one minute on M_2 . Machine M_1 is available for not more than 7 hrs 30 min. while machine M_2 is available for 10 hrs during any working day. Formulate the LPP and find the number of units of products A and B to be manufactured to get maximum profit using graphical method.	5	CO1																																																	
1c	Identify all the basic feasible solutions for the system: $x_1 + 4x_2 + x_3 = 8$ $x_1 + 2x_2 + x_4 = 4$ $x_1, x_2, x_3, x_4 \geq 0.$	5	CO1																																																	
Q2	Attempt any 2 parts of the followings																																																			
2a	Write the dual of the following LPP (Primal) Max $Z = 4x_1 + 3x_2$ subject to $x_1 + x_2 \leq 8$ $2x_1 + x_2 \leq 10$ $x_1, x_2 \geq 0.$ Find the solution of the dual problem by solving the primal.	5	CO2																																																	
2b	We have to assign six jobs, I, II, III, IV, V, VI to six workers A, B, C, D, E, and F. Find the minimum time taken by the workers to complete all jobs. <table><tr><td></td><td>I</td><td>II</td><td>III</td><td>IV</td><td>V</td><td>VI</td></tr><tr><td>A</td><td>5</td><td>0</td><td>6</td><td>8</td><td>7</td><td>4</td></tr><tr><td>B</td><td>5</td><td>2</td><td>3</td><td>0</td><td>6</td><td>6</td></tr><tr><td>C</td><td>3</td><td>4</td><td>4</td><td>3</td><td>5</td><td>2</td></tr><tr><td>D</td><td>3</td><td>4</td><td>9</td><td>7</td><td>2</td><td>3</td></tr><tr><td>E</td><td>0</td><td>1</td><td>3</td><td>2</td><td>1</td><td>2</td></tr><tr><td>F</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td><td>4</td></tr></table>		I	II	III	IV	V	VI	A	5	0	6	8	7	4	B	5	2	3	0	6	6	C	3	4	4	3	5	2	D	3	4	9	7	2	3	E	0	1	3	2	1	2	F	4	3	2	1	0	4	5	CO2
	I	II	III	IV	V	VI																																														
A	5	0	6	8	7	4																																														
B	5	2	3	0	6	6																																														
C	3	4	4	3	5	2																																														
D	3	4	9	7	2	3																																														
E	0	1	3	2	1	2																																														
F	4	3	2	1	0	4																																														
2c	Solve the transportation problem using Vogel's approximation method to find the initial basic feasible solution. Apply UV method to check if the obtained solution is optimal or not. <table><tr><td></td><td>M1</td><td>M2</td><td>M3</td><td>Supply</td></tr><tr><td>W1</td><td>2</td><td>1</td><td>4</td><td>10</td></tr><tr><td>W2</td><td>6</td><td>3</td><td>2</td><td>15</td></tr><tr><td>W3</td><td>4</td><td>2</td><td>3</td><td>20</td></tr><tr><td>Demand</td><td>13</td><td>18</td><td>14</td><td></td></tr></table>		M1	M2	M3	Supply	W1	2	1	4	10	W2	6	3	2	15	W3	4	2	3	20	Demand	13	18	14		5	CO2																								
	M1	M2	M3	Supply																																																
W1	2	1	4	10																																																
W2	6	3	2	15																																																
W3	4	2	3	20																																																
Demand	13	18	14																																																	

Q3	Attempt any 2 parts of the followings																		
3a	Solve the following 2×4 game graphically. <table><tr><td></td><td colspan="4">Player B</td></tr><tr><td></td><td>1</td><td>3</td><td>11</td><td>7</td></tr><tr><td>Player A</td><td>8</td><td>5</td><td>2</td><td>6</td></tr></table>		Player B					1	3	11	7	Player A	8	5	2	6	5	CO3	
	Player B																		
	1	3	11	7															
Player A	8	5	2	6															
3b	Define the two person zero sum game, payoff matrix, and value of the game. For what value of λ , the game with the following payoff matrix is strictly determinable? <table><tr><td></td><td>B_1</td><td>B_2</td><td>B_3</td></tr><tr><td>A_1</td><td>λ</td><td>6</td><td>2</td></tr><tr><td>A_2</td><td>-1</td><td>λ</td><td>-7</td></tr><tr><td>A_3</td><td>-2</td><td>4</td><td>λ</td></tr></table>		B_1	B_2	B_3	A_1	λ	6	2	A_2	-1	λ	-7	A_3	-2	4	λ	5	CO3
	B_1	B_2	B_3																
A_1	λ	6	2																
A_2	-1	λ	-7																
A_3	-2	4	λ																
3c	Define a strategy? What do you mean by pure strategy and mixed strategy? Give examples. Define saddle points. Also describe the Maximin-Minimax principle with example for a two person zero sum game.	5	CO3																
Q4	Attempt any 2 parts of the followings																		
4a	Solve the following mixed integer programming problem: Max $Z = x_1 + x_2$ subject to $2x_1 + 5x_2 \leq 16$ $6x_1 + 5x_2 \leq 30$ $x_1, x_2 \geq 0, x_1$ is an integer.	5	CO4																
4b	Solve the following linear programming problem using dual simplex method. Min $Z = x_1 + 2x_2 + 3x_3 + 4x_4$ subject to $x_1 + 2x_2 + 2x_3 + 3x_4 \geq 30$ $2x_1 + x_2 + 3x_3 + 2x_4 \geq 20$ $x_1, x_2, x_3, x_4 \geq 0$.	5	CO4																
4c	Solve the following mixed integer programming problem: Min $Z = 4x_1 + 3x_2$ subject to $3x_1 + 4x_2 \leq 12$ $4x_1 + 2x_2 \leq 9$ $x_1, x_2 \geq 0$, and are integers.	5	CO4																
Q5	Attempt any 2 parts of the followings																		
5a	Solve the non linear programming problem by forming a bordered hessian matrix: Optimize $Z = x_1^2 + x_2^2 + x_3^2$ subject to $x_1 + x_2 + 3x_3 = 2$ $5x_1 + 2x_2 + x_3 = 5$ $x_1, x_2, x_3 \geq 0$	5	CO5																
5b	Solve the following NLPP: Optimize $Z = 4x_1^2 + 2x_2^2 + x_3^2 - 4x_1x_2$ subject to $x_1 + x_2 + x_3 = 15$ $2x_1 - x_2 + 2x_3 = 20$ $x_1, x_2, x_3 \geq 0$	5	CO5																
5c	Solve the following NLPP using KKT conditions: Optimize $Z = 2x_1 + 3x_2 - (x_1^2 + x_2^2 + x_3^2)$ subject to $x_1 + x_2 \leq 1$ $2x_1 + 3x_2 \leq 6$ $x_1, x_2, x_3 \geq 0$	5	CO5																